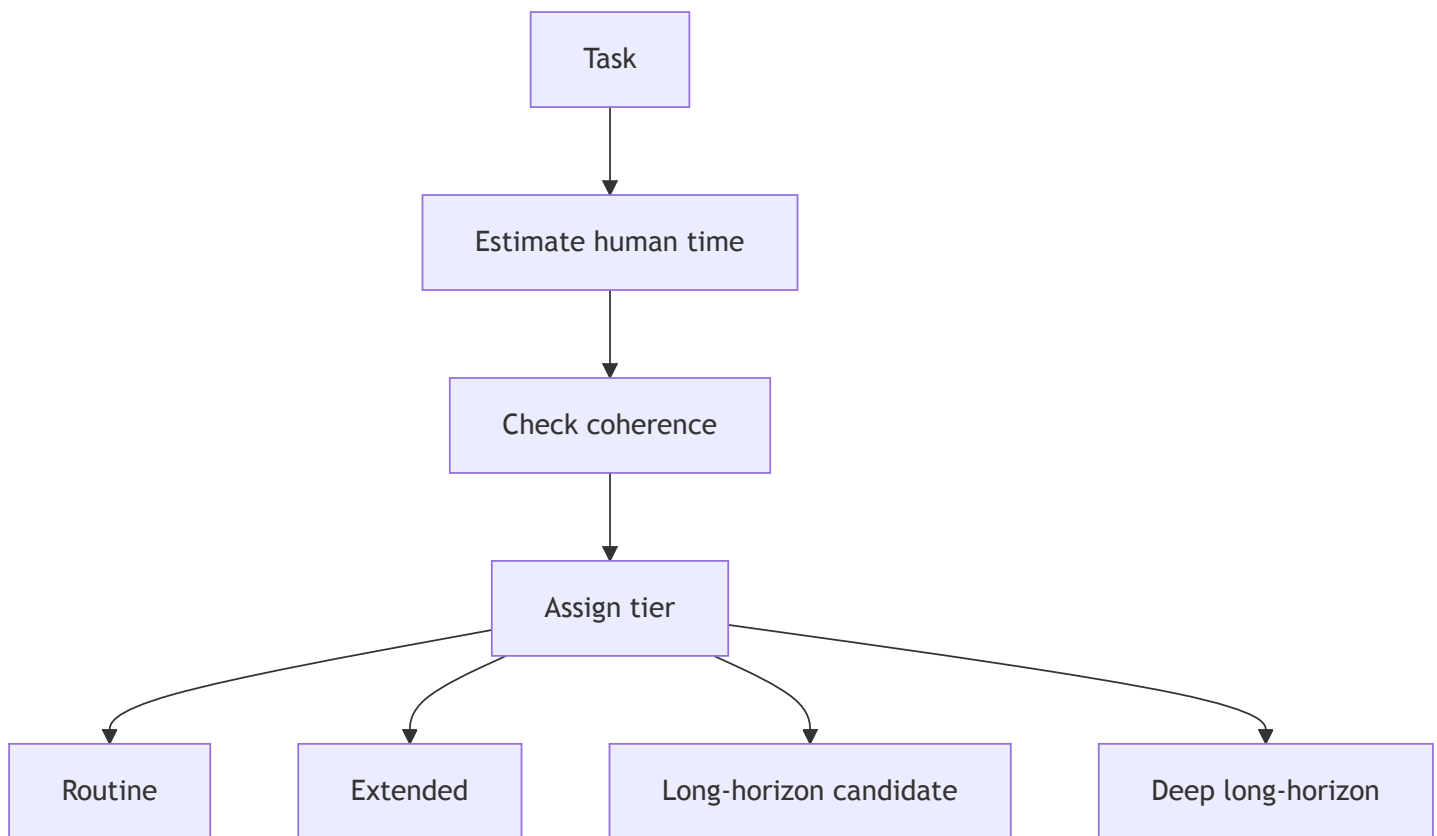


Long-Horizon Task Definition

Part 1: Formal Definition

TL;DR

A long-horizon task is a coherent work objective whose difficulty comes from sustained context intake, decision-making, dependent actions, qualitative judgment, and verification. We estimate it by the time a **competent low-context human** would need using normal workplace interfaces, not by the number of model tool calls.



Human Baseline

Use a **competent low-context human** as the baseline: someone with the relevant professional skill and tool familiarity, but without special prior knowledge of this exact seeded environment, customer, repo, ticket history, or internal shorthand.

This is intentionally stricter than estimating time for the teammate who designed the task, and more realistic than comparing against an LLM that can emit API calls or text faster than a human can read, click, type, and verify in a UI.

Classification Tiers

Estimated human time	Tier	Interpretation
< 1h	Routine	Useful as a smoke/regression task, not long-horizon.
1–4h	Extended	Meaningful workflow, but not enough sustained complexity to call long-horizon.
4–10h	Long-horizon candidate	Plausibly long-horizon if context, decisions, dependencies, and verification are real.
>= 10h	Deep long-horizon	Strong claim: substantial work for a competent low-context human.
Multi-day/week	Frontier productivity simulation	Multiple deliverables, evolving context, or project-scale work.

For stakeholder-facing claims, use **deep long-horizon** for the strongest category. Tasks in the 4–10h band should be treated as candidates until timed review or strong evidence supports the estimate.

Guardrails

Do not classify a task as long-horizon just because it has many tool calls, many rows in a transcript, or repeated similar actions.

Require these checks:

- **Coherent unit:** the work should be one integrated objective, not many independent subtasks glued together.
- **Decision-critical context:** count context that a human must inspect to decide correctly, not every token returned by tools.
- **Non-trivial selection:** plausible distractors, exclusion rules, or target ambiguity should affect the outcome.

- **Dependent actions:** later actions should depend on earlier findings or choices.
- **Qualitative work:** writing, summaries, notes, or handoffs should require judgment, not fixed boilerplate.
- **Verification burden:** the human must confirm both completed changes and avoided side effects.
- **Efficiency realism:** estimate UI/manual human work, not API-call latency or LLM text generation speed.

Examples:

Guardrail	Counts	Does Not Count
Coherent unit	Investigate one customer escalation across tickets, Slack, incidents, and docs, then produce a coordinated handoff.	Ten unrelated tickets that happen to use the same update pattern.
Decision-critical context	Ticket conversations, customer tier, incident history, SLA policy, linked logs, and prior owner notes needed to decide whether a ticket qualifies.	Full JSON payloads, timestamps, IDs, duplicated metadata, or records returned by broad search but never needed for the decision.
Non-trivial selection	Choosing 9 qualifying incidents from 40 plausible candidates using customer impact, platform area, status, and exclusion rules.	Updating every returned record or following an explicit list of target IDs with no ambiguity.
Dependent actions	Reading capacity evidence, choosing owners, updating project notes, scheduling the right review, then notifying the right stakeholders.	Creating three independent calendar events whose details are fully specified.
Qualitative work	Writing distinct handoff notes that explain reason, evidence, and next step for each target.	Copying the same fixed sentence into every record.
Verification burden	Checking changed targets, untouched distractors, recipients, permissions, dates, and side effects.	Confirming only that a command returned success.

Phase-Based Estimator

Use a calibrated human-time estimate by phase:

```
total_minutes =
  context_intake_minutes
+ candidate_screening_minutes
+ decision_planning_minutes
+ execution_minutes
+ qualitative_writing_minutes
+ verification_restraint_minutes
```

Phase	What It Measures
context_intake_minutes	Reading or scanning task instructions, returned records, documents, messages, policies, and relevant state.
candidate_screening_minutes	Comparing candidate items against task rules and excluding distractors.
decision_planning_minutes	Choosing the correct targets, sequencing work, and resolving dependencies.
execution_minutes	Manual UI operations such as edits, record updates, sends, creates, or assignments.
qualitative_writing_minutes	Drafting judgment-heavy notes, emails, summaries, comments, or handoffs.
verification_restraint_minutes	Checking expected changes, no-touch entities, side effects, recipients, timestamps, and final consistency.

Tool calls, context tokens, golden trajectories, and expected diffs are inputs to the estimate. They are evidence, not the definition.

Research Summary

Work	What It Contributes	Takeaway For Our Definition
METR time horizon	Measures model capability by human expert completion time at success thresholds.	Use human-time as the core scale, but treat it as continuous

Work	What It Contributes	Takeaway For Our Definition
		and baseline-dependent, not a universal binary cutoff.
tau-bench	Tests tool-agent-user interaction with domain APIs, policies, and database-state evaluation.	Rule-following and reliable state changes matter more than raw call count.
WorkArena	Evaluates browser agents on enterprise knowledge-work tasks in ServiceNow.	Enterprise tasks require realistic workflow context and multimodal/UI action, but not all are long-horizon.
WorkArena++	Adds compositional planning, retrieval, arithmetic/logical reasoning, and human traces.	Planning and contextual understanding are key dimensions beyond step count.
AgentLongBench	Studies long-context agents through dynamic environment rollouts.	Dense tool-response information can be harder than long chat history; count decision-critical density, not raw tokens alone.
TRIP-Bench	Long-horizon travel-planning benchmark with multi-turn interactions, global constraints, 150+ tool calls, and 200k+ context in hard cases.	Long-horizon work often combines global constraints, changing requirements, and multi-tool coordination.
Odysseys	Realistic long-horizon web tasks from live browsing sessions with graded rubrics and efficiency metrics.	Binary pass/fail is too coarse; long-horizon evaluation should reward partial progress and efficiency.
Synthetic Computers at Scale	Simulates productivity objectives tied to synthetic user computers and multi-deliverable work.	Month-scale productivity work is a different tier from single-session Harbor tasks.
APEX-Agents	Cross-application professional tasks from banking, consulting, and law.	Professional long-horizon tasks should involve realistic files/tools and domain deliverables.

Work	What It Contributes	Takeaway For Our Definition
WildClawBench	Native-runtime tasks averaging more than 20 tool calls and about 8 minutes wall-clock.	A task can be tool-heavy and still short for humans; tool count alone is not sufficient.

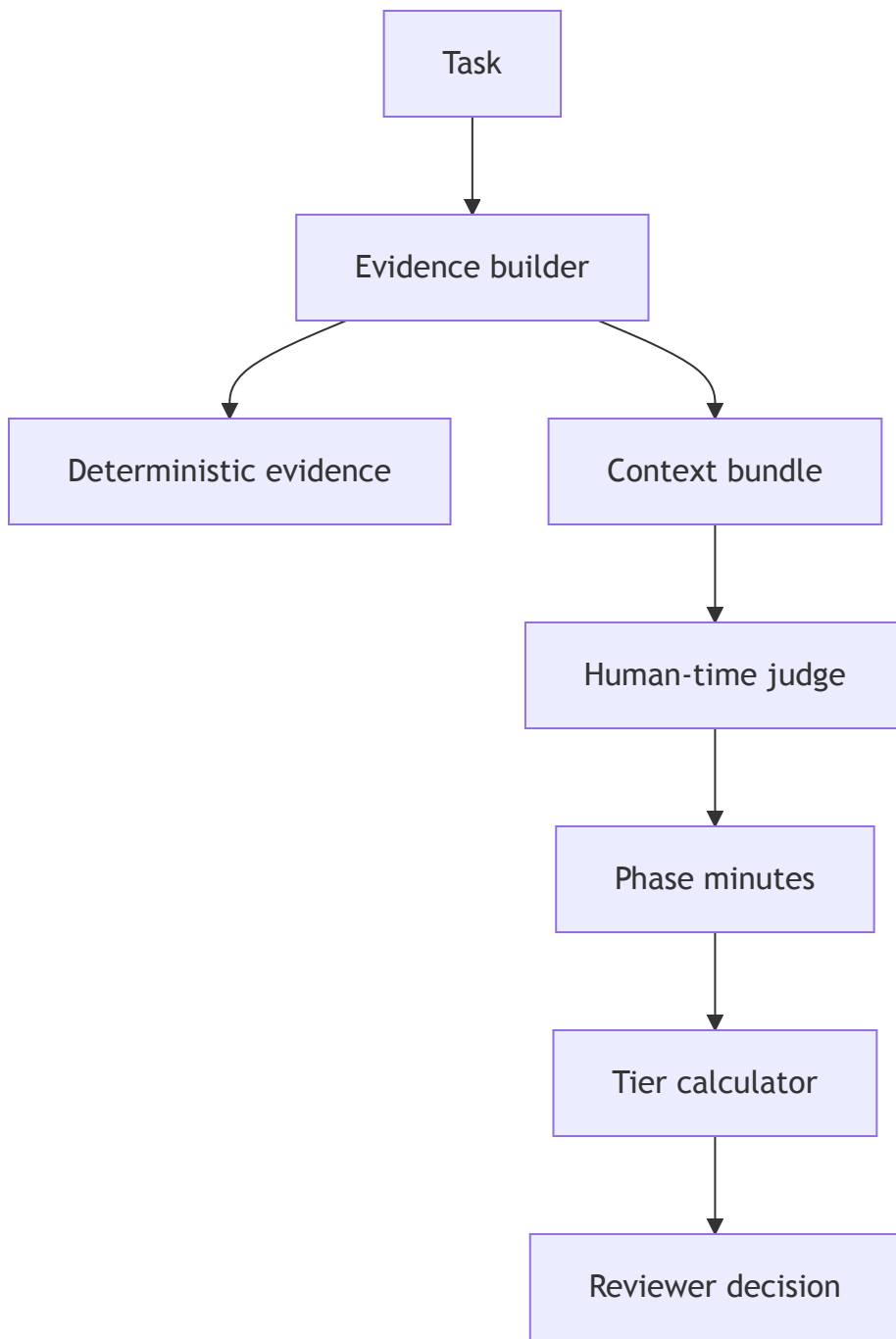
References:

- METR, "Measuring AI Ability to Complete Long Software Tasks": <https://arxiv.org/abs/2503.14499>
- METR time horizon methodology: <https://metr.org/time-horizons/>
- tau-bench: <https://arxiv.org/abs/2406.12045>
- WorkArena: <https://arxiv.org/abs/2403.07718>
- WorkArena++: <https://arxiv.org/abs/2407.05291>
- AgentLongBench: <https://arxiv.org/abs/2601.20730>
- TRIP-Bench: <https://arxiv.org/abs/2602.01675>
- Odysseys: <https://arxiv.org/abs/2604.24964>
- Synthetic Computers at Scale: <https://arxiv.org/abs/2604.28181>
- APEX-Agents: <https://arxiv.org/abs/2601.14242>
- WildClawBench: <https://arxiv.org/abs/2605.10912>

Part 2: Implementation Proposal

Core Proposal

Use a two-stage classifier:



The classifier should be deterministic where possible, LLM-judged where semantic, and tool-using only when static artifacts are insufficient.

Evidence Builder

Inputs:

instruction.md
task.toml
solution/golden_trajectory.json
tests/expected_diff.json
tests/rubric.toml
optional representative read-only context samples

Output:

```
{
  "task_summary": "...",
  "domains": ["freshdesk", "teams", "github"],
  "human_baseline": "competent_low_context_human",
  "tool_calls": [],
  "tool_results": [],
  "token_counts": {},
  "candidate_entities": [],
  "mutation_count": 0,
  "qualitative_outputs": [],
  "rule_evidence": {}
}
```

Variable Extraction

Signal	Best method	Use
Tokens and tool/result counts	Deterministic	Evidence only; never direct classification.
Candidate entities	Deterministic seed plus LLM cleanup	Estimate screening work and distractor burden.
Rule complexity	LLM judge with rubric/instruction evidence	Estimate semantic checks per candidate and planning burden.
Required operations	Deterministic seed plus LLM cleanup	Estimate manual UI/action time.
Qualitative outputs	LLM judge	Estimate writing time by judgment required, not text length alone.

Signal	Best method	Use
Verification burden	Deterministic seed plus LLM cleanup	Estimate target and no-touch checks.

Human-Time Judge

Give the LLM the evidence bundle and ask for structured phase estimates:

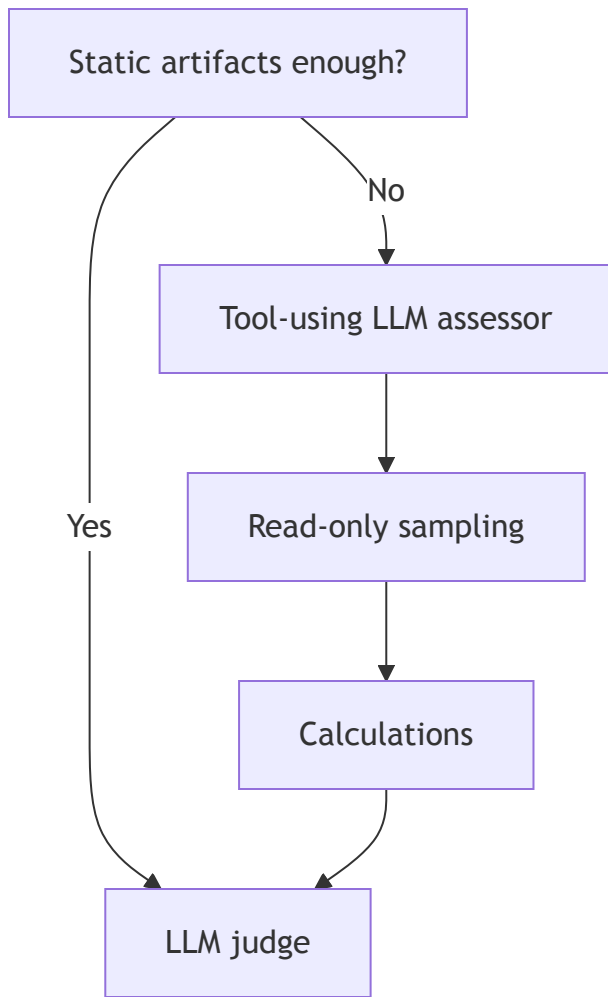
```
{
  "human_baseline": "competent_low_context_human",
  "phase_minutes": {
    "context_intake": 35,
    "candidate_screening": 25,
    "decision_planning": 15,
    "execution": 25,
    "qualitative_writing": 15,
    "verification_restraint": 15
  },
  "total_minutes": 130,
  "tier": "extended",
  "quality_flags": [
    "tool_call_count_is_not_duration",
    "requires_reviewer_calibration"
  ],
  "confidence": 0.7,
  "rationale": "Short explanation grounded in context, decisions, actions, and verifica
}
```

The judge must estimate human UI/manual work. It must not compare the task to model runtime, hidden reasoning tokens, API latency, or LLM text-generation speed.

Tool-Using LLM Assessor

Use a plain LLM judge by default.

Use a tool-using LLM assessor only when the static bundle is insufficient:



This mode is useful when:

- `golden_trajectory.json` has no real tool results;
- the task depends on live discovery;
- multi-domain state is too large to include directly;
- the classifier needs to sample records, files, messages, or other entities before estimating candidate fanout.

The tool-using assessor should use read-only tools and return the same structured estimate as the plain LLM judge.

Review Use

During task review:

1. Build the evidence bundle.
2. Generate a deterministic conservative estimate.
3. Ask an LLM judge for phase minutes when the task is a serious candidate.
4. Compare the output against reviewer intuition and known timed examples.

5. Accept the tier only when the rationale explains real context, decisions, dependencies, qualitative work, and verification.

The classifier is a review aid. Final classification should remain reviewer-approved until enough timed human reviews exist for calibration.